

HW01: Linear Algebra Problem Set

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Instructions. Work neatly and show all steps. You may discuss ideas with a classmate, but write up your own solutions. Scan or photograph your work, combine it into one PDF, name it `lastname_firstname_hw01_problemset.pdf`, and upload it to the HW01 Drive folder (the same one you upload your coding submission to). The problem set is worth **40 points**: 5 for Warmups, 25 for Core Problems, 10 for Challenge Problems.

Remark 1. For extra practice beyond this set, Boyd & Vandenberghe's *Introduction to Applied Linear Algebra* (free PDF online — search “VMLS Boyd Vandenberghe”) has many good applied problems; chapters 1, 3, 6, 8, and 10 match this module.

1 Warmups

Problem 1

Let $u = (3, -1, 2)$ and $v = (1, 4, -2)$.

- (a) Compute $2u - 3v$.
- (b) Compute $u \cdot v$.
- (c) Compute $\|u\|$ and the unit vector pointing in the direction of u .
- (d) Are u and v orthogonal? How can you tell without drawing anything?

2 Core Problems

Problem 2: Orthogonality

- (a) Find the value of x that makes $(x, 2)$ orthogonal to $(3, -1)$.
- (b) Now do it in general: given any nonzero vector (a, b) , write down a formula for a nonzero vector orthogonal to it.
- (c) Geometrically, what did your formula in (b) do to (a, b) ? (Hint: compare the two vectors' lengths and the angle between them.)

Problem 3: Cosine similarity

A search engine represents documents as word-count vectors. Consider

$$d_1 = (4, 0, 2), \quad d_2 = (8, 0, 4), \quad d_3 = (1, 3, 1),$$

and the query $q = (2, 0, 1)$.

- (a) Rank the documents by dot product with q .
- (b) Rank them by cosine similarity with q .
- (c) The two rankings treat d_1 and d_2 differently. What is the relationship between d_1 and d_2 , and which ranking is more sensible for search? Why?

Problem 4: Matrix multiplication

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

- (a) Compute AB and BA . What does multiplying by B on the left do to A 's entries? On the right?
- (b) Conclude: does $AB = BA$ in general?

Problem 5: Transformations compose in order

Let R be rotation by 90° counterclockwise and $S = \begin{bmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$.

- (a) Write down the matrix for R .
- (b) Compute RS and SR .
- (c) Apply both products to the point $(1, 1)$ and sketch the two results. In the product RS , which transformation does the point experience *first*?
- (d) How is this like composing functions $f(g(x))$?

Problem 6: Inverses

(a) Verify by multiplication that $\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$.

(b) Consider the family of matrices

$$C(t) = \begin{bmatrix} 1 & t \\ 2 & 4 \end{bmatrix}.$$

For which real values of t is $C(t)$ invertible? Justify your answer by considering whether its columns are linearly independent.

(c) For a value of t where the inverse fails to exist, find two distinct input points with the same output under $C(t)$. Describe all possible outputs geometrically and explain why the transformation cannot be undone for every input.

Problem 7: Systems of equations

Consider the system

$$2x + y = 5, \quad 5x + 3y = 13.$$

(a) Solve it by hand (substitution or elimination).

(b) Write the system as $Av = b$ for a matrix A and vectors v, b .

(c) You already know A^{-1} from Problem 6. Compute $A^{-1}b$ and confirm it matches your answer from (a).

3 Challenge Problems

Problem 8: Prove it

Let u, v be vectors with n components.

(a) Show that $\|u + v\|^2 = \|u\|^2 + 2u \cdot v + \|v\|^2$. (Expand using the definition of the norm as a dot product.)

(b) For which pairs u, v does $\|u + v\|^2 = \|u\|^2 + \|v\|^2$ hold? What famous theorem did you just recover?

Problem 9: Adding rank-one matrices

Can two rank-one 2×2 matrices have an invertible sum? Construct an example, or explain why it is impossible. Justify the rank of each matrix and the invertibility or noninvertibility of their sum.

Problem 10: Recovering a matrix from two inputs

Let

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

An unknown 2×2 matrix M satisfies

$$Mv_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad Mv_2 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}.$$

- (a) Prove that $\{v_1, v_2\}$ is a linearly independent set, using the definition of linear independence.
- (b) Find Me_1 and Me_2 , where $e_1 = (1, 0)$ and $e_2 = (0, 1)$. Show how linearity lets you obtain these outputs from the two given input-output pairs.
- (c) Write down M and explain how you placed the vectors from (b) into the matrix.
- (d) Check your matrix by multiplying it by both v_1 and v_2 and comparing with the given outputs.

Problem 11: Why networks need more than matrices

In a few weeks you will build neural networks. First consider a simplified two-layer “network” that only multiplies by weight matrices and computes $y = W_2(W_1x)$ for 2×2 weight matrices W_1, W_2 .

- (a) Show that this two-layer network is exactly equivalent to some one-layer network $y = Wx$. What is W ?
- (b) What does this say about stacking ten such layers?
- (c) Now insert an operation between the layers that replaces each negative coordinate with zero. Taking both weight matrices to be the identity gives

$$F(x_1, x_2) = (\max(0, x_1), \max(0, x_2)).$$

For $p = (1, 0)$ and $q = (-1, 0)$, compute $F(p)$, $F(q)$, and $F(p + q)$. Compare $F(p + q)$ with $F(p) + F(q)$. Use your results and the way matrix multiplication treats vector sums to explain why no single matrix can represent F for every input.